

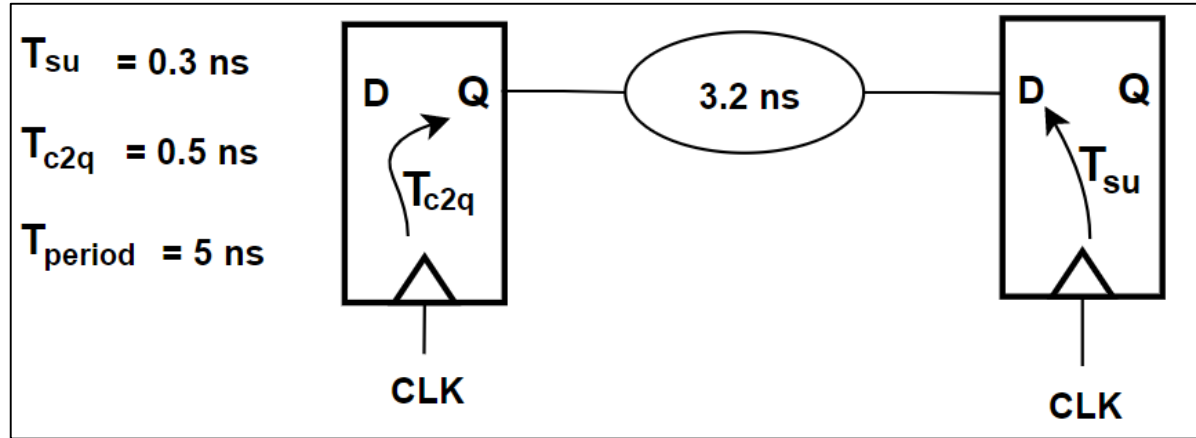
Example 1

- Assumptions

1. Interconnect Delays not considered
2. Clock Uncertainty not considered
3. One Operating Condition

- Required

1. Is this path timing is met or not?
2. if met calculate the positive slack and if not calculate the negative slack



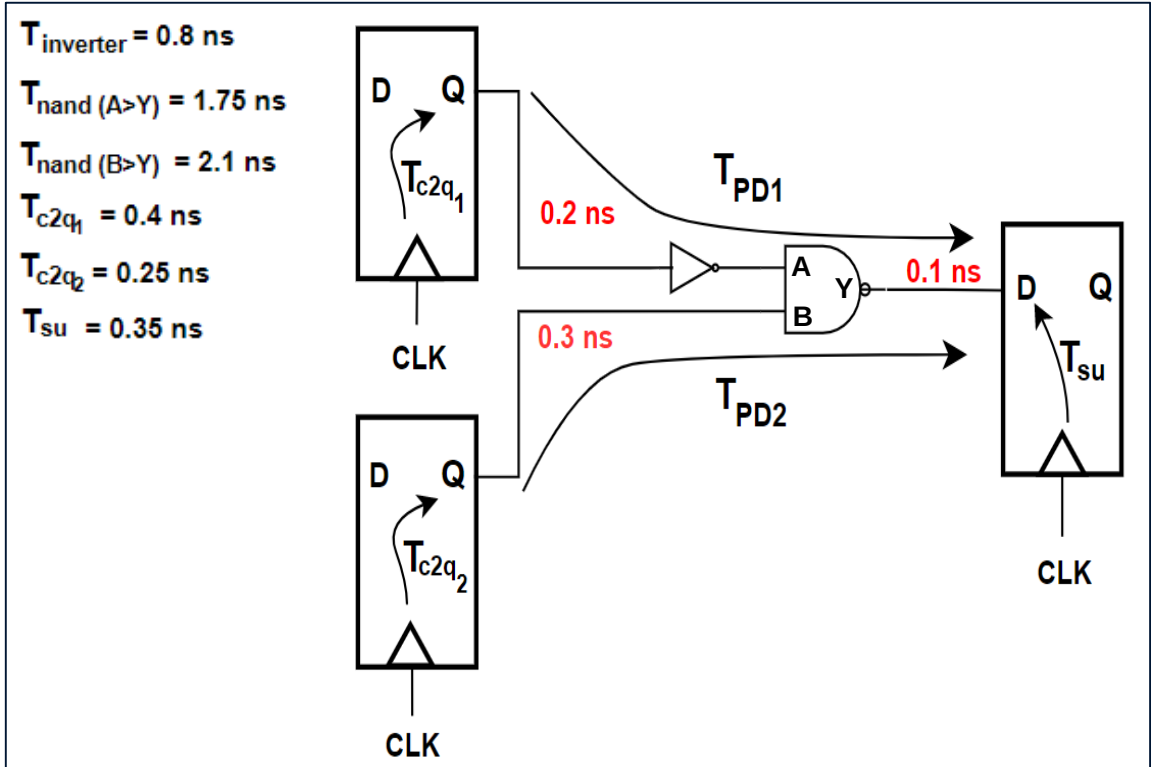
Example 2

- Assumptions

1. **Interconnect Delays is considered**
2. Clock Uncertainty not considered
3. One Operating Condition

Required: -

1. Calculate T_{pd1}
2. Calculate T_{pd2}
3. Calculate the Minimum period (Max Frequency) to avoid setup violation for all paths



Example 3

- Assumptions

1. **Interconnect Delays is considered**
2. **Clock Uncertainty is considered**
3. **One Operating Condition**

Required: -

1. Calculate T_{pd1}
2. Calculate T_{pd2}
3. Calculate the Minimum period (Max Frequency) to avoid setup violation for all paths

$$T_{\text{inverter}} = 0.8 \text{ ns}$$

$$T_{\text{nand}}(A > Y) = 1.75 \text{ ns}$$

$$T_{\text{nand}}(B > Y) = 2.1 \text{ ns}$$

$$T_{c2q_1} = 0.4 \text{ ns}$$

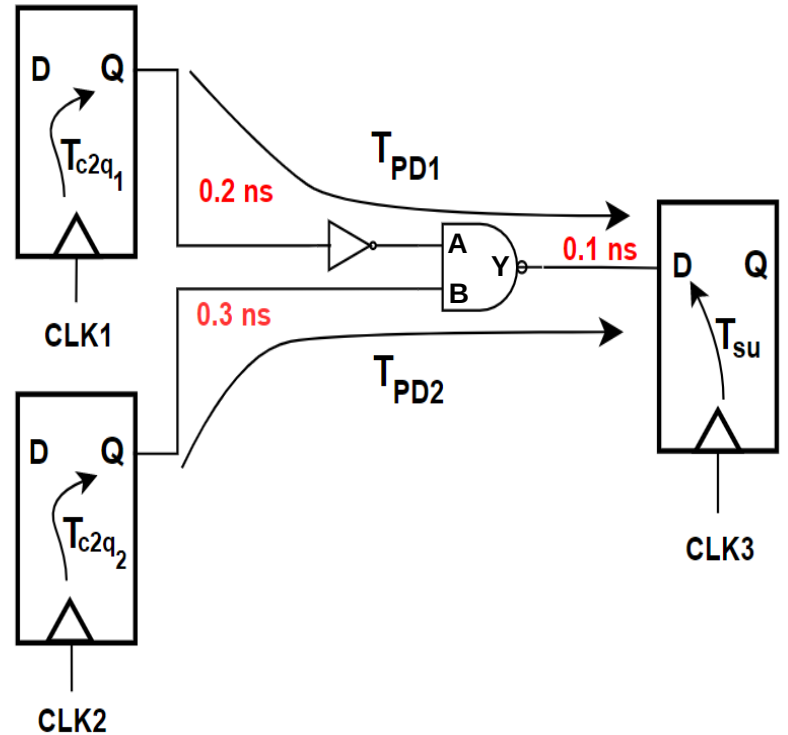
$$T_{c2q_2} = 0.25 \text{ ns}$$

$$T_{\text{su}} = 0.35 \text{ ns}$$

Skew

$$\text{CLK } 1_3 = 0.4 \text{ ns}$$

$$\text{CLK } 2_3 = 0.1 \text{ ns}$$



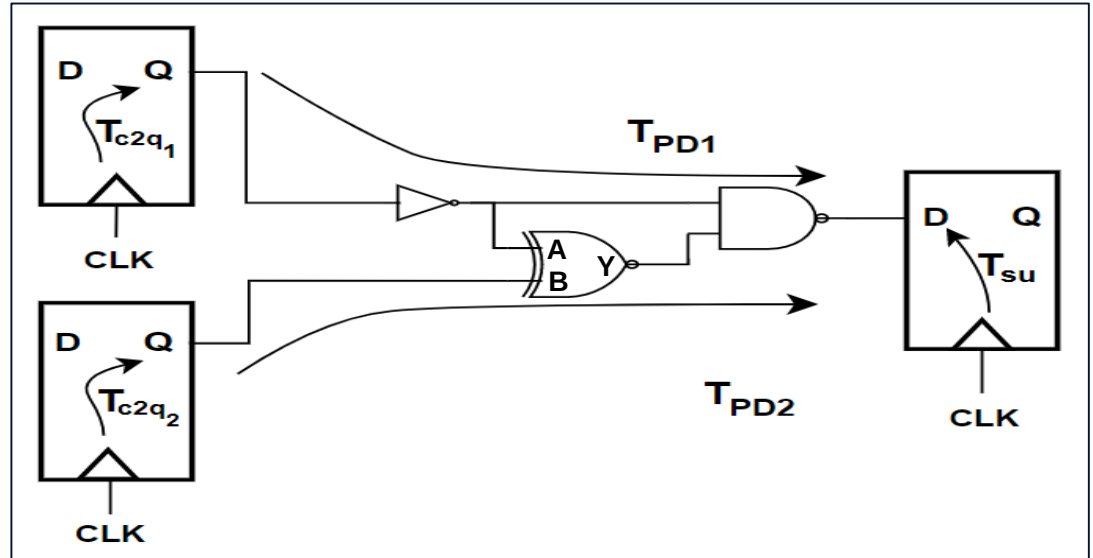
Example 4

- Assumptions**

1. Interconnect Delays is not considered
2. Clock Uncertainty is not considered
3. **Worst/Best Operating Conditions**

Required: -

1. Calculate T_{pd1}
2. Calculate T_{pd2}
3. Calculate the Minimum period (Max Frequency) to avoid setup violation for all paths



Delay	Inverter	XNOR	NAND	T_{cq1}	T_{cq2}	T_{su}
Max	0.8	1.7	1.2	0.4	0.25	0.3
Min	0.65	1.55	1.1	0.3	0.2	0.2